Short Answer Test

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1) How does computer vision enable computers to interpret?

Ans: Computer Vision is a branch of AI that enables computers to capture, process, analyze, and understand images and videos. It uses image processing and machine learning techniques to identify objects, recognize patterns, and make decisions automatically.

(a) Main Objectives :-

→ Extract useful information from images
→ Recognize and classify objects
→ Understand scenes for decision-making.

(b) Two Applications :-

→ Face recognition.
→ Self-driving cars.

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2) Key Features of CV?

Ans Computer Vision extracts meaningful information from images using advanced algorithms. It supports object detection, recognition, tracking, and scene understanding.

(a) Feature extraction: Feature extraction is the process of identifying important image characteristics such as edges, corners, textures and shapes.

(b) Object Recognition: Object recognition identifies and classifies object in an image, improving the accuracy and intelligence of computer Vision Systems.

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3) Different Levels of image processing?

Ans: Computer Vision uses three levels of image processing

- Low Level - Image enhancement and noise removal
- Mid Level - Image segmentation and feature extraction.
- High-Level - Object recognition and scene understanding.

(a) Low-Level - It improves image quality by reducing noise and enhancing brightness or contrast.

(b) Difference - Mid Level, high-Level.

Digital image representation?

Ans:- A digital image consists of pixels arranged in rows and columns. Each Pixel stores intensity or color information.

(a) Pixel: A pixel is the smallest element of a digital image that represents one point of color or intensity.

(b) Effect of resolution:- Higher resolution provides more pixels resulting in clearer and sharper images.

Image sensing and acquistion?

Ans:- Image sensing and acquistion convert real-world scenes into digital images using image sensors and imaging devices.

(a) Role of image sensor: It converts light into electrical signals, which are transformed into digital images

(b) Two acquistion devices:-

→ Digital Camera

→ Scanner.

Sampling and Quantization ?

Sampling and Quantization convert an analog image into a digital image.

a) <u>Sampling</u>: Sampling converts a continuous image into discrete pixels.

(b) <u>Purpose of Quantization</u>: Quantization assigns intensity values to pixels, allowing the image to be stored digitally.

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7) Image Processing Vs Computer Vision ?

<u>Ans</u>: The image formation model explains how light from a scene is captured by a camera to produce an image.

(a) <u>Perspective Projection</u>: It projects a 3D scene onto a 2D image plane, making distant objects appear smaller.

(b) <u>Role of illumination</u>: Illumination provides the light necessary to capture clear and accurate images.

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8) Image Processing ?

<u>Ans</u>: Image Processing improves image quality, while Computer Vision interprets image content for intelligent decision-making.

(a) <u>Image Processing</u>: It performs operations such as filtering, enhancement and restoration of images.

(b) <u>Two differences</u>: Image Processing enhances images; Computer Vision understands images.

- Image Processing outputs an improved image or decision

9) Applications of Computer Vision ?

Ans:- Computer Vision is widely used in health care, transportation, agriculture, robotics and security

(a) Health Care:- It helps detects diseases from x-rays, CT scan, MRI images.

(b) Autonomous Vehicle: It detects roads, traffic signs, pedestrians, and obstacles for safe driving.

10) Digital image fundamentals ?

Ans:- It is such as pixels, sampling, and resolution are essential for accurate image analysis.

(a) Pixels are the basic building blocks of digital images and store color or intensity information.

(b) Higher Sampling and resolution produce more detailed images, image analysis accuracy.